SHIVASUBRAHMANYA K C

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Education

Sahyadri College of Engineering and Management, Mangaluru **2023-2027**

Bachelor of Engineering in Computer Science (Artificial Intelligence & Machine Learning) **CGPA: 9.48**

Research and Innovation

Multimodal Feature Level Fusion for Arecanut Quality Assessment Elsevier *Array* (Submitted); SSRN

Developed a multimodal RGB–X-ray deep learning pipeline for arecanut quality classification, achieving 99.8% accuracy using CNNs with PyTorch and TensorFlow/Keras.

Empowering Wildlife Conservation Through Real-Time Anti-Poaching System [DOI]

Co-authored a real-time IoT and AI based anti-poaching system using YOLOv8n for intrusion detection in wildlife reserves, with SMS-based alerting and explainable AI for forest official use.

Projects

RxSignal - Drug Interaction Risk Intelligence Platform XGBoost, PyTorch, FastAPI, React

- Built an end-to-end drug safety platform ingesting 1.06B FDA adverse event records (FAERS 2025) with a chunked DuckDB pipeline, computing PRR/ROR signals across 43.8M drug-event triples.
- Trained XGBoost (90.6% accuracy, 99.35% recall) and a GraphSAGE GNN on a 13,160-node drug graph; served predictions via FastAPI and React in ∼1ms using pre-computed embeddings.

AESCULAPIUS - Self-Healing Multi-Agent RAG Platform LangGraph, Gemini/Ollama

- Built a self-healing multi-agent RAG platform that retrieves, reranks, verifies, and autonomously improves responses using an 8-agent LangGraph workflow.
- Integrated Pinecone, BAAI embeddings, cross-encoder reranking, hallucination detection, confidence scoring, and FastAPI with a React-based interface.

EEG-Based Early Detection of Neurodegenerative Disorders Python, TensorFlow, React, FastAPI

- Built a Multi-Modal Deep Ensemble using 1D-CNNs and Chebyshev GCNs on 19-channel EEG signals to classify Alzheimer's, FTD, and Cognitively Normal with 85.7% balanced accuracy.
- Developed a 5-seed ensemble combining temporal and spectral features, with 0.960 AUC and 0.853 macro-F1.

Skills

Programming Languages: Python, SQL, Java, C/C++, HTML,CSS, JavaScript

AI/ML & Development: Machine Learning, Deep Learning, Data Visualization, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, React, Node.js, Express.js, MongoDB, LangChain, LangGraph

Tools & Platforms: Git, GitHub, Jupyter, Docker, Vercel, Linux, Postman, n8n, Power BI, Figma, Canva

Soft Skills: Problem-Solving, Strategic Thinking, Collaboration, Leadership, Mentoring

Achievements

RIDA - IEEE YESIST12 Special Track **Finalist**

Developed a GraphSAGE-based clinical decision support system for rare disease diagnosis.

StreamSentinel - IEEE YESIST12 WePOWER Track **Finalist**

Presented a federated IoT framework for real-time water quality monitoring.

VEDHA - VVCE EdTech Hackathon **Finalist**

Built Edu-Ease, an AI-powered assistive learning platform for special-needs children.

Organizations and Volunteering

Chair, IEEE Student Branch & Computational Intelligence Society **2025 – 2026**

Coordinated multiple technical events and workshops, engaging 200+ participants

Hackharbor 3.0 with SOSC

Trained juniors on C, Web Development, APIs, Cloud, LLMs, and GitHub during a 10-day college internship.

Synergia'25 – Documentation Lead

Prepared event documentation, planning records, and post-event summaries in collaboration with multiple teams.

Skill-Link Bridge Course – Session Facilitator

Mentored BCA and BBA students in programming fundamentals, AI tools, and hands-on technical activities.

Code Sprint – Hackathon Evaluator

Assessed hackathon submissions for technical quality, innovation, and solution effectiveness.

Certifications

Google Advanced Data Analytics

Infosys Springboard Machine Learning with TensorFlow

Infosys Springboard Database Management Systems I / II

AWS Fundamentals of Machine Learning & AI

IBM Skillbuild Cloud Computing Fundamentals

API Fundamentals Student Expert